Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version A**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In any vector space V, $0\mathbf{u} = \mathbf{0}$ for every vector $\mathbf{u}$ in V.	☐	☐
(b) The set of all invertible 2×2 matrices is a subspace of M_{22}.	☐	☐
(c) If $S = \{\mathbf{w}_1, \ldots, \mathbf{w}_r\}$ is a set of vectors in a vector space V, then $\text{span}(S)$ is a subspace of V.	☐	☐
(d) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then $\mathbf{v}_3$ must be a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ 0)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$ ☐ B. Axiom 10: $1\mathbf{u} = \mathbf{u}$

☐ C. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ ☐ D. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

(b) Which of the following subsets W is a **subspace** of $V = \mathbb{R}^2$ (with the usual operations)?

☐ A. $W = \{(x, y) : x + y = 1\}$ ☐ B. $W = \{(x, y) : y = x^2\}$

☐ C. $W = \{(x, y) : y = 3x\}$ ☐ D. $W = \{(x, y) : x \geq 0,\ y \geq 0\}$

(c) Let $\mathbf{v}_1 = (1, 0, 2)$ and $\mathbf{v}_2 = (0, 1, -1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

☐ A. $(3, 1, 7)$ ☐ B. $(3, 1, 5)$ ☐ C. $(1, 1, 2)$ ☐ D. $(2, 2, 0)$

(d) Let $f_1 = x$ and $f_2 = e^x$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = (x - 1)e^x$; the set is linearly dependent because $W(1) = 0$ ☐ B. $W(x) = (x - 1)e^x$; the set is linearly independent

☐ C. $W(x) = (x + 1)e^x$; the set is linearly independent ☐ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 2, -1), \qquad \mathbf{v}_2 = (2, 1, 1), \qquad \mathbf{v}_3 = (5, 4, 1),$$

and let $\mathbf{u} = (1, 1, 3)$.

(a) **[2.5 marks]** Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) **[1.5 marks]** Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version B**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) A set with two operations that satisfies nine of the ten vector space axioms is still a vector space.	☐	☐
(b) If A is an $m \times n$ matrix, then the solution set of the homogeneous system $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{R}^n$.	☐	☐
(c) $\text{span}\{1,\ x,\ x^2\} = P_3$.	☐	☐
(d) Every set of four vectors in $\mathbb{R}^3$ is linearly dependent.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 10: $1\mathbf{u} = \mathbf{u}$

☐ B. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

☐ C. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ D. Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

(b) Which of the following subsets of $V = M_{22}$ (with the usual operations) is **not** a subspace of V?

☐ A. all symmetric matrices ($A^T = A$)

☐ B. all invertible matrices

☐ C. all matrices with $\text{tr}(A) = 0$

☐ D. all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $a = 0$

(c) Let $p_1 = 1 + x^2$ and $p_2 = x$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $3 + 2x - 3x^2$ ☐ B. $2 + 3x + 3x^2$ ☐ C. $3 + 2x + 3x^2$ ☐ D. $3 + 3x + 2x^2$

(d) Let $f_1 = e^x$ and $f_2 = e^{2x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 3e^{3x}$; the set is linearly independent

☐ B. $W(x) = 0$ for all x; the Wronskian test is inconclusive

☐ C. $W(x) = e^{3x}$; the set is linearly dependent because e^{2x} is a multiple of e^x

☐ D. $W(x) = e^{3x}$; the set is linearly independent

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x + x^2, \qquad p_2 = 1 - x + 2x^2, \qquad p_3 = 2 + x - x^2,$$

and let $u = 3 - 2x + 2x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version C**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) If $\mathbf{u}$ is a vector in a vector space and k is a scalar with $k\mathbf{u} = \mathbf{0}$, then $k = 0$ or $\mathbf{u} = \mathbf{0}$.	☐	☐
(b) If W_1 and W_2 are subspaces of a vector space V, then $W_1 \cup W_2$ is always a subspace of V.	☐	☐
(c) The vectors $(1, 0, 0)$ and $(0, 1, 0)$ span $\mathbb{R}^3$.	☐	☐
(d) A set with exactly two vectors is linearly independent if and only if neither vector is a scalar multiple of the other.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (k^2 x_1,\ k^2 x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ ☐ B. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

☐ C. Axiom 10: $1\mathbf{u} = \mathbf{u}$ ☐ D. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

(b) Which of the following subsets W is a **subspace** of $V = P_2$ (with the usual operations)?

☐ A. $W = \{p \in P_2 : \deg(p) = 2\}$ ☐ B. $W = \{a + bx + cx^2 : a, b, c \text{ are integers}\}$

☐ C. $W = \{p \in P_2 : p(0) = 1\}$ ☐ D. $W = \{a + bx + cx^2 : a + b + c = 0\}$

(c) Let $\mathbf{v}_1 = (1, 0, -1)$ and $\mathbf{v}_2 = (0, 1, 3)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

☐ A. $(2, 1, 1)$ ☐ B. $(2, 1, 5)$ ☐ C. $(1, 2, -5)$ ☐ D. $(2, 1, -1)$

(d) Let $f_1 = 1$ and $f_2 = e^{-x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = -e^{-x}$; the set is linearly independent ☐ B. $W(x) = e^{-x}$; the set is linearly independent

☐ C. $W(x) = -e^{-x}$; the set is linearly dependent because $W(x) < 0$ ☐ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 1, 2), \qquad \mathbf{v}_2 = (1, 0, 1), \qquad \mathbf{v}_3 = (2, 1, 1),$$

and let $\mathbf{u} = (3, 3, 4)$.

(a) **[2.5 marks]** Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) **[1.5 marks]** Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version D**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a vector space, if $k\mathbf{u} = \mathbf{0}$ for a scalar k and a vector $\mathbf{u}$, then both $k = 0$ and $\mathbf{u} = \mathbf{0}$.	☐	☐
(b) If W_1 and W_2 are subspaces of a vector space V, then $W_1 \cap W_2$ is a subspace of V.	☐	☐
(c) Every vector in a set S belongs to $\text{span}(S)$.	☐	☐
(d) The functions e^x and $6e^x$ form a linearly independent set in $C(-\infty, \infty)$.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (0,\ kx_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

☐ B. Axiom 4: there is a zero vector $\mathbf{0}$ with $\mathbf{0} + \mathbf{u} = \mathbf{u}$

☐ C. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

☐ D. Axiom 10: $1\mathbf{u} = \mathbf{u}$

(b) Which of the following subsets of $V = \mathbb{R}^3$ (with the usual operations) is **not** a subspace of V?

☐ A. $W = \{(x, y, z) : x + y + z = 1\}$

☐ B. $W = \{(x, y, z) : x + y + z = 0\}$

☐ C. $W = \{(x, y, z) : x = y = z\}$

☐ D. $W = \{(x, y, z) : x = 2y \text{ and } z = 0\}$

(c) Let $p_1 = 1 + x$ and $p_2 = x^2$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $2 - 2x - x^2$ ☐ B. $2 + x - x^2$ ☐ C. $1 + 2x + 2x^2$ ☐ D. $2 + 2x - x^2$

(d) Let $f_1 = x$ and $f_2 = x^2$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = x^2$; the set is linearly dependent because $W(0) = 0$

☐ B. $W(x) = x^2$; the Wronskian test is inconclusive because $W(0) = 0$

☐ C. $W(x) = x^2$; the set is linearly independent

☐ D. $W(x) = 3x^2$; the set is linearly independent

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x, \qquad p_2 = 1 - x + x^2, \qquad p_3 = 1 + 3x - x^2,$$

and let $u = 1 + 2x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version E**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) The set $F(-\infty,\infty)$ of all real-valued functions defined on $(-\infty,\infty)$, with pointwise addition and scalar multiplication, is a vector space.	☐	☐
(b) The set of all $n \times n$ matrices A with $\operatorname{tr}(A) = 0$ is a subspace of M_{nn}.	☐	☐
(c) If $\operatorname{span}(S) = \mathbb{R}^3$, then S must contain exactly three vectors.	☐	☐
(d) If the Wronskian $W(x)$ of two functions satisfies $W(0) = 0$, then the Wronskian test is inconclusive.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1,x_2,x_3)+(y_1,y_2,y_3) = (x_1+y_1,\ x_2+y_2,\ x_3+y_3)$ and the scalar multiplication $k(x_1,x_2,x_3) = (kx_1,\ x_2,\ x_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$ ☐ B. Axiom 8: $(k+m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ C. Axiom 10: $1\mathbf{u} = \mathbf{u}$ ☐ D. Axiom 2: $\mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u}$

(b) Which of the following subsets W is a **subspace** of $V = F(-\infty,\infty)$ (with the usual operations)?

☐ A. $W = \{f : f(0) = 1\}$ ☐ B. $W = \{f : f(x) \geq 0 \text{ for all } x\}$

☐ C. $W = C(-\infty,\infty)$, all continuous functions ☐ D. $W = \{f : f(1) = 2\}$

(c) Let $\mathbf{v}_1 = (1,1,0)$ and $\mathbf{v}_2 = (0,1,1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\operatorname{span}\{\mathbf{v}_1,\mathbf{v}_2\}$?

☐ A. $(2,3,5)$ ☐ B. $(2,5,3)$ ☐ C. $(2,5,4)$ ☐ D. $(1,1,1)$

(d) Let $f_1 = e^x$ and $f_2 = xe^x$ be functions in $C^\infty(-\infty,\infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = e^{2x}$; the set is linearly independent ☐ B. $W(x) = xe^{2x}$; the set is linearly independent

☐ C. $W(x) = xe^{2x}$; the set is linearly dependent because $W(0) = 0$ ☐ D. $W(x) = e^{2x}$; the set is linearly dependent because xe^x is a multiple of e^x

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, -1, 2), \qquad \mathbf{v}_2 = (1, 1, 0), \qquad \mathbf{v}_3 = (1, -3, 4),$$

and let $\mathbf{u} = (1, 2, 1)$.

(a) **[2.5 marks]** Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) **[1.5 marks]** Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version F**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) To show that a set with two operations is a vector space, it is enough to verify Axiom 1 (closure under addition) and Axiom 6 (closure under scalar multiplication).	☐	☐
(b) The first quadrant $W = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0\}$ is a subspace of $\mathbb{R}^2$.	☐	☐
(c) $\text{span}\{1, x, x^2\} = P_2$.	☐	☐
(d) If the Wronskian of $f_1, \ldots, f_n$ is not identically zero on $(-\infty, \infty)$, then the functions are linearly independent.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1, x_2, x_3) + (y_1, y_2, y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1, x_2, x_3) = (kx_1,\ kx_2,\ 0)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 1: $\mathbf{u} + \mathbf{v}$ is in V (closure under addition)

☐ B. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ C. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

☐ D. Axiom 10: $1\mathbf{u} = \mathbf{u}$

(b) Which of the following subsets W is a **subspace** of $V = M_{22}$ (with the usual operations)?

☐ A. all matrices with $\det(A) = 0$

☐ B. all matrices with $\text{tr}(A) = 0$

☐ C. all matrices of the form $\begin{bmatrix} 1 & a \\ b & c \end{bmatrix}$

☐ D. all matrices whose four entries are all ≥ 0

(c) Let $p_1 = 1 - x$ and $p_2 = 1 + x^2$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $1 + 2x - 3x^2$ ☐ B. $3 + 2x + x^2$ ☐ C. $1 + 2x + 3x^2$ ☐ D. $1 - 2x + 3x^2$

(d) Let $f_1 = 1$ and $f_2 = x^2$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 2x$; the set is linearly dependent because $W(0) = 0$

☐ B. $W(x) = -2x$; the set is linearly independent

☐ C. $W(x) = 2x$; the Wronskian test is inconclusive because $W(0) = 0$

☐ D. $W(x) = 2x$; the set is linearly independent

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + x^2, \qquad p_2 = x + x^2, \qquad p_3 = 1 - x + x^2,$$

and let $u = -1 + 4x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version G**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a vector space V, $(-1)\mathbf{u} = -\mathbf{u}$ for every vector $\mathbf{u}$ in V.	☐	☐
(b) The set $W = \{a + bx + cx^2 : a,\ b,\ c \text{ are integers}\}$ is a subspace of P_2.	☐	☐
(c) The zero vector belongs to $\text{span}(S)$ for every nonempty set S of vectors in V.	☐	☐
(d) If the Wronskian of $f_1, \ldots, f_n$ is identically zero on $(-\infty, \infty)$, then the functions are linearly dependent.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (kx_1,\ k^2x_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 10: $1\mathbf{u} = \mathbf{u}$ ☐ B. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

☐ C. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ ☐ D. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$

(b) Which of the following subsets of $V = \mathbb{R}^2$ (with the usual operations) is **not** a subspace of V?

☐ A. $W = \{(x, y) : x = 0\}$ (the y-axis) ☐ B. $W = \{(x, y) : 2x - 5y = 0\}$

☐ C. $W = \{(0, 0)\}$ ☐ D. $W = \{(x, y) : xy \geq 0\}$

(c) Let $\mathbf{v}_1 = (1, 0, 0)$ and $\mathbf{v}_2 = (0, 1, 2)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

☐ A. $(4, 2, 1)$ ☐ B. $(1, 2, 3)$ ☐ C. $(0, 1, 3)$ ☐ D. $(4, 1, 2)$

(d) Let $f_1 = e^{2x}$ and $f_2 = e^{3x}$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 5e^{5x}$; the set is linearly independent ☐ B. $W(x) = e^{5x}$; the set is linearly independent

☐ C. $W(x) = e^{5x}$; the set is linearly dependent because $e^{3x} = e^x \cdot e^{2x}$ ☐ D. $W(x) = e^{6x}$; the set is linearly independent

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, 2, 1), \qquad \mathbf{v}_2 = (2, 1, -1), \qquad \mathbf{v}_3 = (1, 1, 2),$$

and let $\mathbf{u} = (4, 3, -3)$.

(a) **[2.5 marks]** Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) **[1.5 marks]** Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Working:

Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version H**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) If $\mathbf{u}$ is a vector in a vector space and k is a scalar with $k\mathbf{u} = \mathbf{0}$, then necessarily $k = 0$.	☐	☐
(b) The set $W = \{a + bx + cx^2 : a + b + c = 0\}$ is a subspace of P_2.	☐	☐
(c) The set $S = \{(1,-1,0),\ (1,2,1),\ (1,0,1),\ (3,3,-1)\}$ cannot span $\mathbb{R}^3$ because it contains more than three vectors.	☐	☐
(d) Every set of five matrices in M_{22} is linearly dependent.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1,x_2,x_3)+(y_1,y_2,y_3) = (x_1+y_1,\ x_2+y_2,\ x_3+y_3)$ and the scalar multiplication $k(x_1,x_2,x_3) = (0,\ kx_2,\ kx_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 10: $1\mathbf{u} = \mathbf{u}$

☐ B. Axiom 8: $(k+m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ C. Axiom 4: there is a zero vector $\mathbf{0}$ with $\mathbf{0} + \mathbf{u} = \mathbf{u}$

☐ D. Axiom 1: $\mathbf{u} + \mathbf{v}$ is in V (closure under addition)

(b) Which of the following subsets of $V = F(-\infty,\infty)$ (with the usual operations) is **not** a subspace of V?

☐ A. $W = \{f : f(0) = 1\}$

☐ B. $W = \{f : f(0) = 0\}$

☐ C. $W = C^1(-\infty,\infty)$, all continuously differentiable functions

☐ D. $W = P_2$, all polynomials of degree at most 2

(c) Let $p_1 = x + x^2$ and $p_2 = 1$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $5 + 2x + 2x^2$ ☐ B. $5 + 2x + 5x^2$ ☐ C. $2 + 5x + 2x^2$ ☐ D. $5 + 2x - 2x^2$

(d) Let $f_1 = x$ and $f_2 = 1 + x$ be functions in $C^\infty(-\infty,\infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 1$; the set is linearly independent

☐ B. $W(x) = -1$; the set is linearly dependent because $W(x)$ is negative

☐ C. $W(x) = -1$; the set is linearly independent

☐ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 2 + x - x^2, \qquad p_2 = 1 + x + x^2, \qquad p_3 = 4 + 3x + x^2,$$

and let $u = 1 + 2x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

Working:

Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version I**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a vector space V, the equation $k\mathbf{0} = \mathbf{0}$ holds only when the scalar k equals 0.	☐	☐
(b) The set $C(-\infty, \infty)$ of all continuous functions on $(-\infty, \infty)$ is a subspace of $F(-\infty, \infty)$.	☐	☐
(c) If $\mathbf{u} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, then $\mathbf{u} = k_1\mathbf{v}_1 + k_2\mathbf{v}_2$ for some scalars k_1, k_2.	☐	☐
(d) Every set of two nonzero vectors in $\mathbb{R}^3$ is linearly independent.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^2$ with the usual addition $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1,\ x_2 + y_2)$ and the scalar multiplication $k(x_1, x_2) = (x_1,\ kx_2)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 7: $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$ ☐ B. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

☐ C. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ ☐ D. Axiom 10: $1\mathbf{u} = \mathbf{u}$

(b) Which of the following subsets W is a **subspace** of $V = \mathbb{R}^3$ (with the usual operations)?

☐ A. $W = \{(x, y, z) : x \geq 0\}$ ☐ B. $W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$

☐ C. $W = \{(x, y, z) : xyz = 0\}$ ☐ D. $W = \{(x, y, z) : z = x - 2y\}$

(c) Let $\mathbf{v}_1 = (1, 0, 1)$ and $\mathbf{v}_2 = (0, 1, 1)$ in $\mathbb{R}^3$. Which of the following vectors belongs to $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$?

☐ A. $(3, -1, 4)$ ☐ B. $(3, 1, 2)$ ☐ C. $(3, -1, 2)$ ☐ D. $(1, 1, 1)$

(d) Let $f_1 = x^2$ and $f_2 = x^3$ be functions in $C^\infty(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = x^4$; the set is linearly dependent because $W(0) = 0$ ☐ B. $W(x) = x^4$; the set is linearly independent

☐ C. $W(x) = 5x^4$; the set is linearly independent ☐ D. $W(x) = 0$ for all x; the Wronskian test is inconclusive

Problem 3. Linear independence and span [4 marks]

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be the set of vectors in $\mathbb{R}^3$ given by

$$\mathbf{v}_1 = (1, -2, 1), \qquad \mathbf{v}_2 = (3, 1, 2), \qquad \mathbf{v}_3 = (5, -3, 4),$$

and let $\mathbf{u} = (2, 1, 3)$.

(a) **[2.5 marks]** Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) **[1.5 marks]** Determine whether $\mathbf{u}$ belongs to $\text{span}(S)$, that is, whether $\mathbf{u}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. If it is, write $\mathbf{u}$ as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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Qatar University

Department of Mathematics, Statistics and Physics

Department of Mathematics and Statistics

Math 231 – Linear Algebra Quiz 3

Sections 4.1–4.4 (Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence)

Fall 2026 • Thursday, 5 November 2026 • Time: 20 minutes • Total: 10 marks

Name: _____ ID: _____ Section: _____ **Version J**

Tick exactly one box for each True/False statement and exactly one option for each multiple-choice question; an item with more than one answer receives zero. Show all your work in Problem 3. Programmable calculators and mobile phones are not allowed.

Problem 1. True or False [2 marks]

Each statement is worth 0.5 mark. Tick **True** if the statement is always true, otherwise tick **False**. No justification is needed.

	True	False
(a) In a real vector space, the scalars may be any real numbers, including irrational numbers such as $\sqrt{2}$.	☐	☐
(b) If A is an $m \times n$ matrix and $\mathbf{b} \neq \mathbf{0}$, then the solution set of $A\mathbf{x} = \mathbf{b}$ is a subspace of $\mathbb{R}^n$.	☐	☐
(c) The polynomial x^2 belongs to $\text{span}\{1 + x^2,\ x\}$.	☐	☐
(d) A set consisting of a single nonzero vector is linearly independent.	☐	☐

Problem 2. Multiple choice [4 marks]

Each question is worth 1 mark. Tick exactly one option.

(a) Let $V = \mathbb{R}^3$ with the usual addition $(x_1, x_2, x_3) + (y_1, y_2, y_3) = (x_1 + y_1,\ x_2 + y_2,\ x_3 + y_3)$ and the scalar multiplication $k(x_1, x_2, x_3) = (x_1,\ x_2,\ kx_3)$. With these operations V is *not* a vector space: exactly one of the following four axioms fails. Which one?

☐ A. Axiom 6: $k\mathbf{u}$ is in V (closure under scalar multiplication)

☐ B. Axiom 8: $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$

☐ C. Axiom 10: $1\mathbf{u} = \mathbf{u}$

☐ D. Axiom 9: $k(m\mathbf{u}) = (km)\mathbf{u}$

(b) Which of the following subsets of $V = P_2$ (with the usual operations) is **not** a subspace of V?

☐ A. $W = \{p \in P_2 : p(0) = 0\}$

☐ B. $W = \{a + bx + cx^2 : a + b + c = 2\}$

☐ C. $W = \{a + bx + cx^2 : a = 2c\}$

☐ D. $W = \{p \in P_2 : p(1) = 0\}$

(c) Let $p_1 = 1 + 2x$ and $p_2 = x^2$ in P_2. Which of the following polynomials belongs to $\text{span}\{p_1, p_2\}$?

☐ A. $2 + 2x - x^2$ ☐ B. $2 + 4x - x^2$ ☐ C. $4 + 2x - x^2$ ☐ D. $1 + 4x - x^2$

(d) Let $f_1 = e^{-x}$ and $f_2 = e^{x}$ be functions in $C^{\infty}(-\infty, \infty)$. Compute the Wronskian $W(x)$ of f_1, f_2 and choose the correct statement.

☐ A. $W(x) = 2$; the set is linearly independent

☐ B. $W(x) = 0$ for all x; the Wronskian test is inconclusive

☐ C. $W(x) = -2$; the set is linearly independent

☐ D. $W(x) = 2$; the set is linearly dependent because $W(x)$ is constant

Problem 3. Linear independence and span [4 marks]

Let $S = \{p_1, p_2, p_3\}$ be the set of polynomials in P_2 given by

$$p_1 = 1 + 2x, \qquad p_2 = x - x^2, \qquad p_3 = 1 + x^2,$$

and let $u = 1 - x + x^2$.

(a) [2.5 marks] Determine whether S is linearly independent or linearly dependent. Show all your work.

(b) [1.5 marks] Determine whether u belongs to span(S), that is, whether u is a linear combination of p_1, p_2, p_3. If it is, write u as such a linear combination; if it is not, explain why. (You may reuse the row operations from part (a).)

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